

Demonstration of Proposed Features

Date	2 July, 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features proposed during the planning phase and verifies whether they were successfully implemented and demonstrated in the OptiCrop system for intelligent agricultural decision-making.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Soil & Environmental Data Input	Accepts inputs like Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall through a user-friendly interface	Implemented	Yes	Successfully captures all required inputs
2	Data Preprocessing	Cleans dataset, handles missing values, and normalizes environmental parameters before prediction	Implemented	Yes	Improves prediction accuracy
3	Crop Recommendation Model	Predicts the most suitable crop based on given soil and environmental conditions	Implemented	Yes	ML model integrated successfully.
4	Prediction Result Display	Displays recommended crop and suitability insights instantly	Implemented	Yes	Real-time output generated
5	Model Performance Evaluation	Evaluates model using metrics such as accuracy and validation techniques	Implemented	Yes	Ensures reliability of predictions

6	Flask Web Application Deployment	Integrates ML model into a Flask-based web application for real-time usage	Implemented	Yes	End-to-end system successfully demonstrated
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Feature Implementation Summary

Feature	Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%